

SIMATS ENGINEERING
ASSESSMENT TOOL 1
COURSE NAME: Artificial Intelligence
COURSE CODE: CSA17

Code of Conduct:

I, **B.Vishnu / 192512410** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Annexure A
CASE STUDY SOLUTIONS

Case Study 1: Smart Medical Diagnosis System

(a) Propositional Logic Representation

We define the propositional symbols representing the facts and conclusions:

- *F*: Patient has Fever
- *C*: Patient has Cough
- *R*: Patient has Rash
- *B*: Patient has Breathlessness
- *Flu*: Possible Flu
- *M*: Possible Measles
- *P*: Possible Pneumonia

Logical Rules:

- **Rule I:** $F \wedge C \rightarrow Flu$
- **Rule II:** $F \wedge R \rightarrow M$
- **Rule III:** $C \wedge B \rightarrow P$

Facts: $F = True, C = True, B = True$

(b) Forward Chaining

Forward chaining starts with the known facts and applies rules to derive new conclusions until no new rules can be fired.

- **Iteration 1:** Initial Knowledge Base (KB) contains $\{ F, C, B \}$.
- **Step 1:** Evaluate Rule I ($F \wedge C \rightarrow Flu$). Since F and C are both True in the KB, the rule fires.
Derived Fact: $Flu = True$. Update KB: $\{ F, C, B, Flu \}$.
- **Step 2:** Evaluate Rule II ($F \wedge R \rightarrow M$). Since R is unknown (not in KB), the rule does not fire.
- **Step 3:** Evaluate Rule III ($C \wedge B \rightarrow P$). Since C and B are both True in the KB, the rule fires.
Derived Fact: $P = True$. Update KB: $\{ F, C, B, Flu, P \}$.

Conclusion: The possible derived diagnoses for Patient A are **Flu** and **Pneumonia**.

(c) Backward Chaining

The goal is to prove P ('Patient A has Pneumonia'). Backward chaining starts with the goal and works backward to see if it is supported by the facts.

- **Goal:** P
- **Step 1:** Find a rule with P as the consequent. Rule III ($C \wedge B \rightarrow P$) matches this.
- **Step 2:** The new sub-goals are C and B . Both must be proven True to satisfy the rule.
- **Step 3:** Check KB for sub-goal C . It is given as a True fact.
- **Step 4:** Check KB for sub-goal B . It is given as a True fact.

Goal Reduction Tree:

- P
 - $\searrow$ Rule III requires C AND B
 - $\rightarrow C$ (Fact: True)
 - $\rightarrow B$ (Fact: True)

Conclusion: Since all sub-goals resolve to True facts, the goal P is successfully verified.

Case Study 2: Autonomous Traffic Management Agent

(a) Unification

- **Rule 1:** $\forall x: Vehicle(x) \wedge EmergencyType(x) \rightarrow ClearPath(x)$
- **Given Facts:** $Vehicle(Ambulance), EmergencyType(Ambulance)$
- **Step 1:** Match the predicate $Vehicle(x)$ from the rule with the fact $Vehicle(Ambulance)$. Substitution $\theta_1 = \{x / Ambulance\}$.
- **Step 2:** Apply θ_1 to the next predicate: $EmergencyType(Ambulance)$. Match this with the fact $EmergencyType(Ambulance)$. No new substitutions are needed.
- **Result:** The complete substitution set (Most General Unifier) is $\theta = \{x / Ambulance\}$.

(b) Resolution Method

To prove $Proceed(CarA)$ using resolution, we must first convert the rules into Conjunctive Normal Form (CNF) and negate the goal.

CNF Conversion:

1. **Rule 1:** $\neg Vehicle(x) \vee \neg EmergencyType(x) \vee ClearPath(x)$
2. **Rule 2:** $\neg ClearPath(x) \vee (GreenSignal(x) \wedge \neg RedSignal(x))$. This splits into two clauses:
 - 2a: $\neg ClearPath(x) \vee GreenSignal(x)$
 - 2b: $\neg ClearPath(x) \vee \neg RedSignal(x)$
3. **Rule 3:** $\neg Vehicle(x) \vee \neg Behind(x,y) \vee \neg GreenSignal(y) \vee Proceed(x)$
4. **Facts:** $Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)$.
5. **Negated Goal:** $\neg Proceed(CarA)$

Resolution Steps:

- **Step 1:** Resolve Rule 1 with facts $Vehicle(Ambulance)$ and $EmergencyType(Ambulance)$ using substitution $\{x/Ambulance\}$.
Derived Clause 6: $ClearPath(Ambulance)$
- **Step 2:** Resolve Clause 2a with Clause 6 using $\{x/Ambulance\}$.
Derived Clause 7: $GreenSignal(Ambulance)$
- **Step 3:** Resolve Rule 3 with facts $Vehicle(CarA), Behind(CarA, Ambulance)$, and Clause 7 ($GreenSignal(Ambulance)$) using $\{x/CarA, y/Ambulance\}$.
Derived Clause 8: $Proceed(CarA)$
- **Step 4:** Resolve Clause 8 with the Negated Goal ($\neg Proceed(CarA)$).
Result: Empty Clause ($\emptyset$). Contradiction found, proving the original goal $Proceed(CarA)$ is True.

(c) Analysis for Simultaneous Emergencies

The current First-Order Logic (FOL) knowledge base is **insufficient** for handling two simultaneous emergency vehicles.

- **Logical Flaw:** If two ambulances approach the same intersection from conflicting directions, Rule 1 and Rule 2 will trigger for both, issuing a $GreenSignal(x)$ to both vehicles. The KB lacks spatial awareness and intersection logic.
- **Additional Rules Needed:**
 - **Intersection Coordinates/Proximity:** Rules defining $Approaching(x, IntersectionID)$ to track where vehicles are.
 - **Mutual Exclusion:** A rule preventing conflicting green lights at the same junction (e.g., $\forall x \forall y: (Crossing(x,y) \wedge GreenSignal(x)) \rightarrow \neg GreenSignal(y)$).

Case Study 3: Agricultural Expert Reasoning System

(a) CNF Conversion

We eliminate implications ($A \rightarrow B \equiv \neg A \vee B$) and apply De Morgan's laws where necessary.

- **P1:** $SoilDry \rightarrow IrrigationNeeded$
CNF: $\neg SoilDry \vee IrrigationNeeded$
- **P2:** $IrrigationNeeded \wedge CropWheat \rightarrow ApplyDripMethod$
Eliminate Implication: $\neg (IrrigationNeeded \wedge CropWheat) \vee ApplyDripMethod$
Move Negation Inward (De Morgan's): $\neg IrrigationNeeded \vee \neg CropWheat \vee ApplyDripMethod$
- **P3:** $ApplyDripMethod \wedge CropWheat \rightarrow CropAtRisk$
Eliminate Implication: $\neg (ApplyDripMethod \wedge CropWheat) \vee CropAtRisk$
Move Negation Inward (De Morgan's): $\neg ApplyDripMethod \vee \neg CropWheat \vee CropAtRisk$
- **Facts:** $SoilDry$ (True), $CropWheat$ (True)

(b) Resolution Refutation Proof

- **Negated Goal:** $\neg ApplyDripMethod$
- **Proof Steps:**
 1. Resolve P1 ($\neg SoilDry \vee IrrigationNeeded$) with the fact $SoilDry$.
Result: $IrrigationNeeded$
 2. Resolve P2 ($\neg IrrigationNeeded \vee \neg CropWheat \vee ApplyDripMethod$) with the fact $CropWheat$.
Result: $\neg IrrigationNeeded \vee ApplyDripMethod$
 3. Resolve the result of Step 1 ($IrrigationNeeded$) with the result of Step 2 ($\neg IrrigationNeeded \vee ApplyDripMethod$).
Result: $ApplyDripMethod$
 4. Resolve $ApplyDripMethod$ with the Negated Goal ($\neg ApplyDripMethod$).
Result: Empty Clause ($\emptyset$). The theorem is proved.

(c) Analysis of Removing 'SoilDry'

If the fact $SoilDry$ is removed from the KB:

- P1 ($SoilDry \rightarrow IrrigationNeeded$) can no longer trigger. We cannot conclude $IrrigationNeeded$.
- Without $IrrigationNeeded$, P2 cannot trigger because the conjunction ($IrrigationNeeded \wedge CropWheat$) is false/unknown. We cannot conclude $ApplyDripMethod$.
- Without $ApplyDripMethod$, P3 cannot trigger because its antecedent condition is incomplete.

Conclusion: The conclusion 'CropAtRisk' **no longer logically follows**. This represents a form of *monotonic logic failure* where the chain of forward inference is broken by the absence of a root premise.

Case Study 4: Wumpus World Knowledge Agent

(a) Belief State & Modus Ponens

The agent processes its percepts using Modus Ponens ($A, A \rightarrow B \vdash B$):

- **Percept 1:** *Stench* at [1,2].
Rule 1: $\text{Stench}[1,2] \rightarrow \text{Wumpus is adjacent to cell [1,2]}$.
Modus Ponens Derived Fact: Wumpus is adjacent to [1,2].
- **Percept 2:** *Breeze* at [1,1].
Rule 2: $\text{Breeze}[1,1] \rightarrow \text{Pit is adjacent to cell [1,1]}$.
Modus Ponens Derived Fact: Pit is adjacent to [1,1].
- **Percept 3:** *Glitter* at [2,2].
Rule 3: $\text{Glitter}[x,y] \rightarrow \text{Gold is at [x,y]}$.
Modus Ponens Derived Fact: Gold is at [2,2].

(b) Forward Chaining for Locations

Using the derived facts from (a), we map out the logical adjacency based on a standard grid (up, down, left, right):

- **Wumpus Location:** The Wumpus is adjacent to [1,2].
Adjacent cells to [1,2] are [1,1], [1,3], and [2,2].
Possible Wumpus Locations: [1,1], [1,3], or [2,2].
- **Pit Location:** The Pit is adjacent to [1,1].
Adjacent cells to [1,1] are [1,2] and [2,1].
Possible Pit Locations: [1,2] or [2,1].

(c) Path Safety Evaluation ([1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2])

We evaluate the safety using Rule 4. (Note: Assuming the standard logical intent of Rule 4 is $\neg \text{Wumpus}[x,y] \wedge \neg \text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$).

- **Cell [1,1]:** The agent is already at [1,1], making it by default a safe square without a Wumpus or Pit.
- **Cell [1,2]:** From part (b), the Pit is at [1,2] or [2,1]. Because we cannot logically prove that [1,2] does NOT contain a Pit ($\neg \text{Pit}[1,2]$ is unproven), the cell [1,2] is marked as **unsafe** (a potential Pit location).

Conclusion: The path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is **NOT safe** because the agent risks falling into a Pit by stepping into cell [1,2]. The agent must explore elsewhere (like [2,1]) to gather more percepts and rule out [1,2] before proceeding.